# FOR A DEGREE P PURELY INSEPARABLE STEP OVER A WEAKLY UNRAMIFIED EXTENSION ONE OF THREE EXPLICITLY DESCRIBED ALTERNATIVES HOLDS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The author writes '(computation omitted)' inside the counted proof span, reproduced verbatim. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

*Allowed standard prerequisites (editorial).*

- Ramification indices of a tower of discrete valuation rings and their multiplicativity, with the ramification index of a purely inseparable degree p step being 1 or p. used at proof lines 34698-34701. Reduces the case analysis to e = p and gives the agreement of the residue fields.
- Weakly unramified extensions and their behaviour under a totally ramified base change. the hypothesis at statement line 34682 and used at proof lines 34718-34719. Is what is preserved when passing to K_1 and B_1, and is one of the conclusions in the third alternative.
- The degree-q extension construction with prescribed uniformizer relation. invoked at proof lines 34712-34716. Produces the auxiliary K_1 and the uniformizer pi' used in the computation.
- Nagata rings and the descent of p-th roots from the completion. the hypothesis at statement line 34681 and used at proof lines 34740-34742. Is what settles the second case.

*Source reference.* Original lemma 09E2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11885–11885. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 32936: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32936–32936. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 34641: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34641–34641. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 34663: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34663–34663. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 32965: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32965–32965. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 34490: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34490–34490. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $A \subset B \subset C$ be extensions of discrete valuation rings with fractions fields $K \subset L \subset M$. Let $\pi \in A$ be a uniformizer. Assume*

1. *$B$ is a Nagata ring,*
2. *$A \subset B$ is weakly unramified,*
3. *$M$ is a degree $p$ purely inseparable extension of $L$.*

*Then either*

1. *$A \to C$ is weakly unramified, or*
2. *$C = B[\pi^{1/p}]$, or*
3. *there exists a degree $p$ separable extension $K_1/K$ totally ramified with respect to $A$ such that $L_1 = L \otimes_K K_1$ and $M_1 = M \otimes_K K_1$ are fields and the maps of integral closures $A_1 \to B_1 \to C_1$ are weakly unramified extensions of discrete valuation rings.*

*Proof.* Let $e$ be the ramification index of $C$ over $B$. If $e = 1$, then we are done. If not, then $e = p$ by Lemmas 32936 and 32965. This in turn implies that the residue fields of $B$ and $C$ agree. Choose a uniformizer $\pi_C$ of $C$. Write $\pi_C^p = u\pi$ for some unit $u$ of $C$. Since $\pi_C^p \in L$, we see that $u \in B^*$. Also $M = L[\pi_C]$.

Suppose there exists an integer $m \geq 0$ such that

$$u = \sum\nolimits_{0 \leq i < m} b_i^p \pi^i + b\pi^m$$

with $b_i \in B$ and with $b \in B$ an element whose image in $\kappa_B$ is not a $p$th power. Choose an extension $K_1/K$ as in Lemma 34641 with $n = m + 2$ and denote $\pi'$ the uniformizer of the integral closure $A_1$ of $A$ in $K_1$ such that $\pi = (\pi')^p + (\pi')^{np} a$ for some $a \in A_1$. Let $B_1$ be the integral closure of $B$ in $L \otimes_K K_1$. Observe that $A_1 \to B_1$ is weakly unramified by Lemma 34490. In $B_1$ we have

$$u\pi = \Big(\sum\nolimits_{0 \leq i < m} b_i (\pi')^{i+1}\Big)^p + b(\pi')^{(m+1)p} + (\pi')^{np} b_1$$

for some $b_1 \in B_1$ (computation omitted). We conclude that $M_1$ is obtained from $L_1$ by adjoining a $p$th root of

$$b + (\pi')^{n-m-1} b_1$$

Since the residue field of $B_1$ equals the residue field of $B$ we see from Lemma 34663 that $M_1/L_1$ has degree $p$ and the integral closure $C_1$ of $B_1$ is weakly unramified over $B_1$. Thus we conclude in this case.

If there does not exist an integer $m$ as in the preceding paragraph, then $u$ is a $p$th power in the $\pi$-adic completion of $B_1$. Since $B$ is Nagata, this means that $u$ is a $p$th power in $B_1$ by Algebra, Lemma 09E2. Whence the second case of the statement of the lemma holds. $\square$